Project-Euler Problem 26

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1 Problem Description

A unit fraction contains 1 in the numerator. The decimal representation of the unit fractions with denominators 2 to 10 are given:

$$\frac{1}{2} = 0.5 \tag{1}$$
$$\frac{1}{3} = 0.(3) \tag{2}$$
$$\frac{1}{4} = 0.25 \tag{3}$$
$$\frac{1}{5} = 0.2 \tag{4}$$
$$\frac{1}{6} = 0.1(6) \tag{5}$$
$$\frac{1}{7} = 0.(142857) \tag{6}$$
$$\frac{1}{8} = 0.125 \tag{7}$$
$$\frac{1}{9} = 0.(1) \tag{8}$$
$$\frac{1}{10} = 0.1 \tag{9}$$

Where $0.1(6)$ means $0.166666\cdots$, and has a 1-digit recurring cycle. It can be seen that $\frac{1}{7}$ has a 6-digit recurring cycle. Find the value of $d < 1000$ for which $\frac{1}{d}$ contains the longest recurring cycle in its decimal fraction part.

2 Algebraic Formulation and the Coprime Kernel

The decimal expansion of a rational number $\frac{1}{d}$ is generated by the execution of the division algorithm in base 10. This process structurally corresponds to evaluating the orbit of the initial state under a shift map determined by multiplication by 10 inside the residue class ring $\mathbb{Z}/d\mathbb{Z}$.

To isolate the purely periodic component of the decimal expansion, the denominator $d \in \mathbb{Z}^+$ must be decomposed via its unique prime factorization. Let:

$$d = 2^a \cdot 5^b \cdot d' \tag{10}$$

where $a, b \in \mathbb{Z}_{\geq 0}$ and $\gcd(d', 10) = 1$. The component $2^a \cdot 5^b$ governs the non-periodic terminating prefix (the transient phase of the orbit), as 2 and 5 are the prime divisors of the base radix 10. The integer d' constitutes the *coprime kernel* of the denominator. The fraction $\frac{1}{d}$ can be expanded as:

$$\frac{1}{d} = \frac{1}{2^a \cdot 5^b \cdot d'} = \frac{1}{10^{\max(a,b)}} \cdot \frac{10^{\max(a,b)}}{2^a \cdot 5^b \cdot d'} \tag{11}$$

By invoking the Bezout identity, this splits into a terminating decimal and a purely periodic fractional component $\frac{c}{d'}$. Thus, the length of the repetend of $\frac{1}{d}$ is identically equal to the length of the repetend of $\frac{1}{d'}$.

2.1 The Multiplicative Group of Units and Period Length

For the coprime kernel d', since $\gcd(10, d') = 1$, the residue class $10 \pmod{d'}$ is an element of the multiplicative group of units modulo d', denoted by:

$$U(d') = (\mathbb{Z}/d'\mathbb{Z})^\times = \{x \in \mathbb{Z}/d'\mathbb{Z} \mid \gcd(x, d') = 1\} \tag{12}$$

The algebraic structure $(U(d'), \cdot)$ forms a finite abelian group whose order (cardinality) is precisely given by Euler's totient function, $|U(d')| = \phi(d')$.

The period length $\lambda(d)$ of the repeating decimal configuration is the minimal size of the cyclic subgroup $\langle 10 \rangle$ generated by 10 within $U(d')$. In number theory, this fundamental invariant is defined as the *multiplicative order* of 10 modulo d':

$$\lambda(d) = \mathrm{ord}_{d'}(10) = \min\{k \in \mathbb{Z}^+ \mid 10^k \equiv 1 \pmod{d'}\} \tag{13}$$

2.2 Fundamental Theorems Governing the Repetend Length

Theorem 1 (Euler's Totient Theorem)**.** *If* $\gcd(10, d') = 1$*, then the order of the group upper-bounds the cyclic trajectory of any of its units:*

$$10^{\phi(d')} \equiv 1 \pmod{d'} \tag{14}$$

Proof. Consider the elements of the group of units $U(d') = \{x_1, x_2, \dots, x_{\phi(d')}\}$. Because multiplication by 10 is an automorphism on $U(d')$, the map $x_i \mapsto 10x_i \pmod{d'}$ merely permutes the elements of the group. Taking the product of all elements yields:

$$\prod_{i=1}^{\phi(d')} x_i \equiv \prod_{i=1}^{\phi(d')} (10x_i) \equiv 10^{\phi(d')} \left(\prod_{i=1}^{\phi(d')} x_i \right) \pmod{d'} \tag{15}$$

Since each x_i is coprime to d', their product is also coprime to d' and possesses a multiplicative inverse. Multiplying both sides by $\left(\prod x_i\right)^{-1}$ isolates the identity:

$$10^{\phi(d')} \equiv 1 \pmod{d'} \tag{16}$$

□

While Euler's theorem establishes that $\phi(d')$ is a valid cycle length, it does not guarantee minimality. To constrain the search space of the minimal exponent $\lambda(d)$, we introduce Lagrange's foundational theorem from modern algebra.

Theorem 2 (Lagrange's Subgroup Theorem)**.** *If H is a subgroup of a finite group G, then the order of H divides the order of G ($|H| \mid |G|$).*

Applied directly to our system, the cyclic group $\langle 10 \rangle = \{10^1, 10^2, \dots, 10^{\lambda(d)}\}$ forms a legitimate subgroup of $U(d')$. Therefore, the order of the element must divide the order of the group:

$$\lambda(d) \;\Big|\; \phi(d') \tag{17}$$

This condition heavily restricts the domain of possible repetend lengths. Instead of executing a sequential linear search across all integers up to $\phi(d')$, the true period length $\lambda(d)$ is guaranteed to sit inside the discrete set of algebraic divisors of $\phi(d')$:

$$\lambda(d) \in \mathcal{D}(\phi(d')) = \{k \in \mathbb{Z}^+ \mid \phi(d') \pmod{k} = 0\} \tag{18}$$

3 Computational Solution and Optimizations

The brute-force approach to finding $\lambda(d)$ simulates long division by tracking remainders linearly, requiring $\mathcal{O}(d)$ time and space complexities. For large scales, this becomes a critical performance bottleneck. By exploiting the properties of the group of units $(\mathbb{Z}/d'\mathbb{Z})^\times$ derived in the mathematical background, we implement an optimized algorithm structured around four main high-performance phases.

3.1 Algorithmic Pipeline

The computational algorithm pipeline translates our group-theoretic constraints into a fast, four-step runtime routine:

1. **Kernel Isolation:** Strip factors of 2 and 5 via continuous division until $d \not\equiv 0 \pmod 2$ and $d \not\equiv 0 \pmod 5$. This isolates the coprime kernel d' in $\mathcal{O}(\log d)$ time. If $d' = 1$, the fraction terminates, and the function returns 0 immediately.

2. **Totient Computation:** Compute $\phi(d')$ by mapping the prime factorization of the kernel to Euler's product formula.

3. **Divisor Sieve:** Extract and sort all elements of the discrete divisor set $\mathcal{D}(\phi(d'))$.

4. **Lagrange Verification:** Traverse $\mathcal{D}(\phi(d'))$ in strictly ascending order. The first divisor k fulfilling $10^k \equiv 1 \pmod{d'}$ is returned as the exact minimal period length.

3.2 Complexity and Performance Optimization Tricks

To maximize throughput and prevent runtime exceptions, three vital optimization strategies are embedded into the engine:

3.2.1 Square-Root Factorization Bounds

Both Step 2 (Euler's Totient) and Step 3 (Divisor Sieve) avoid expensive linear loops by utilizing the algebraic property that non-trivial factors must exist below the square root of the target value. By looping up to $\sqrt{d'}$, the prime factorization of the kernel completes in $\mathcal{O}(\sqrt{d'})$ cycles. Similarly, divisors always emerge in complementary pairs $(i, \frac{\phi(d')}{i})$. Sifting up to $\sqrt{\phi(d')}$ collects all candidate periods dynamically, reducing space and time complexities of divisor generation to $\mathcal{O}(\sqrt{\phi(d')})$.

3.2.2 Binary Exponentiation and Overflow Prevention

Evaluating the modular congruence $10^k \pmod{d'}$ directly is computationally impossible for large k due to structural integer overflow. To bypass this, we utilize *Exponentiation by Squaring* (Binary Exponentiation). This reduces the complexity of computing the modular power from a linear $\mathcal{O}(k)$ to a logarithmic $\mathcal{O}(\log k)$ operations.

To guarantee complete safety against temporary arithmetic overflows during the intermediate squaring steps, the multiplication phase utilizes native compiler-level 128-bit integers via an explicit assignment cast. This allows the engine to evaluate massive exponent boundaries within safe hardware-register thresholds without the computational overhead of arbitrary-precision BigInt libraries.

3.2.3 Search Execution Complexity Profile

Combining these optimizations transforms the total runtime profile per denominator from an unoptimized $\mathcal{O}(d)$ down to a localized bound of:

$$\mathcal{O}\left(\sqrt{d'} + \sqrt{\phi(d')} + |\mathcal{D}(\phi(d'))| \log(\phi(d'))\right) \tag{19}$$

This allows the search loop for all denominators $d < 1000$ to complete in less than 2 milliseconds.

4 Result

Running the optimized algorithm for all $d < 1000$ yields:

$$\boxed{d = 983 \text{ with period length } \lambda = 982} \tag{20}$$

The denominator 983 is prime, and since 10 is a primitive root modulo 983, the period length achieves the maximum possible value of $\phi(983) = 982$.

Most sources where found here: Repeating Decimal